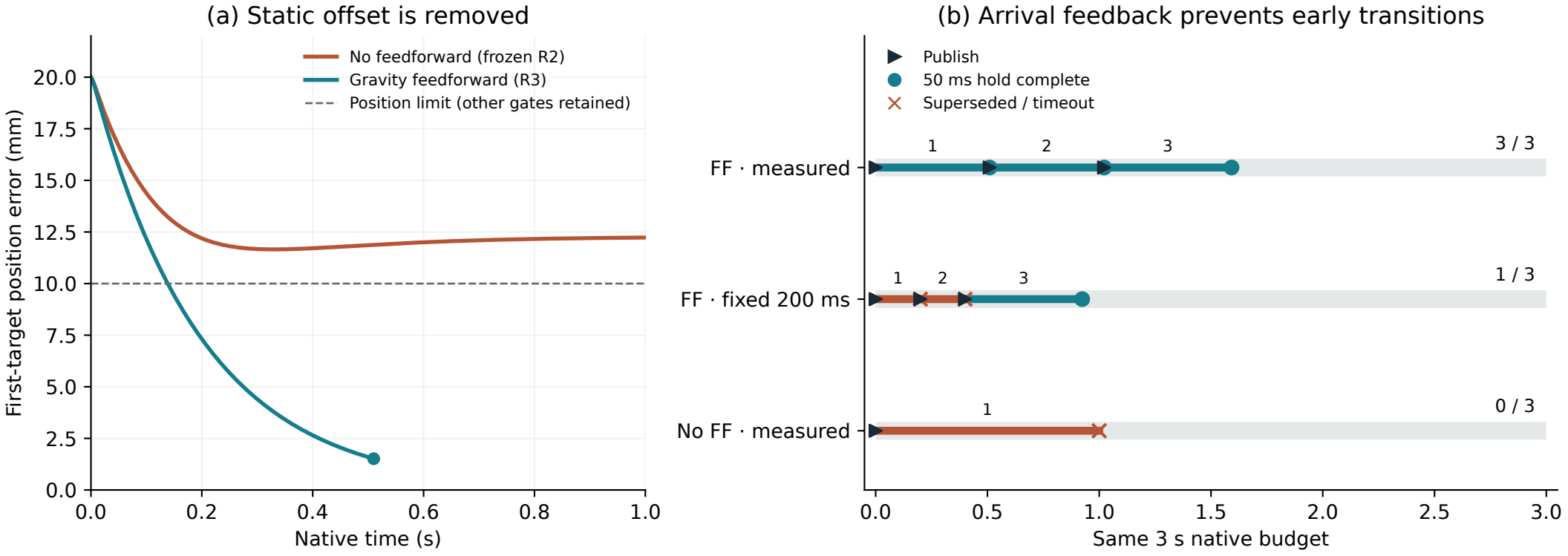
(a) Static offset is removed
No feedforward (frozen R2)
Gravity feedforward (R3)
Position limit (other gates retained)
First-target position error (mm)
Native time (s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
0.0
0.2
0.4
0.6
0.8
1.0
(b) Arrival feedback prevents early transitions
Publish
50 ms hold complete
Superseded / timeout
FF · measured
FF · fixed 200 ms
No FF · measured
1
2
3
3 / 3
1
2
3
1 / 3
1
0 / 3
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Same 3 s native budget